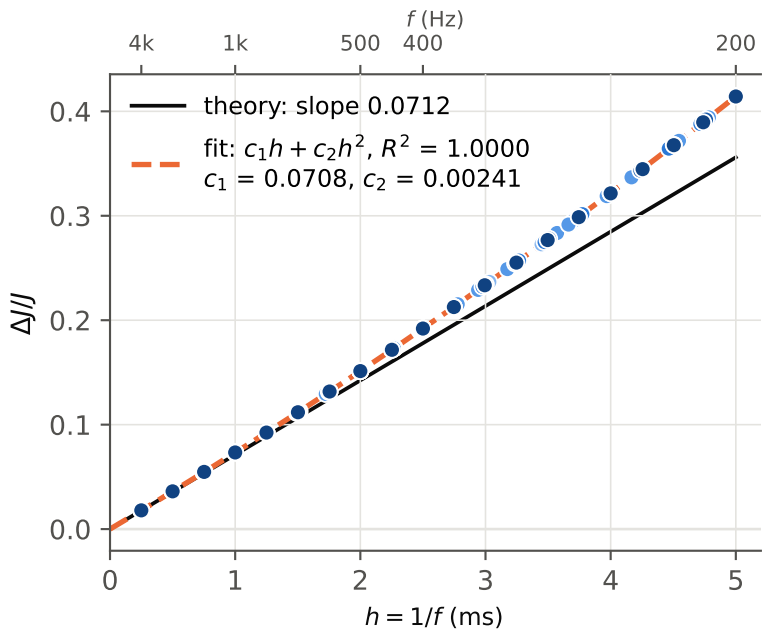
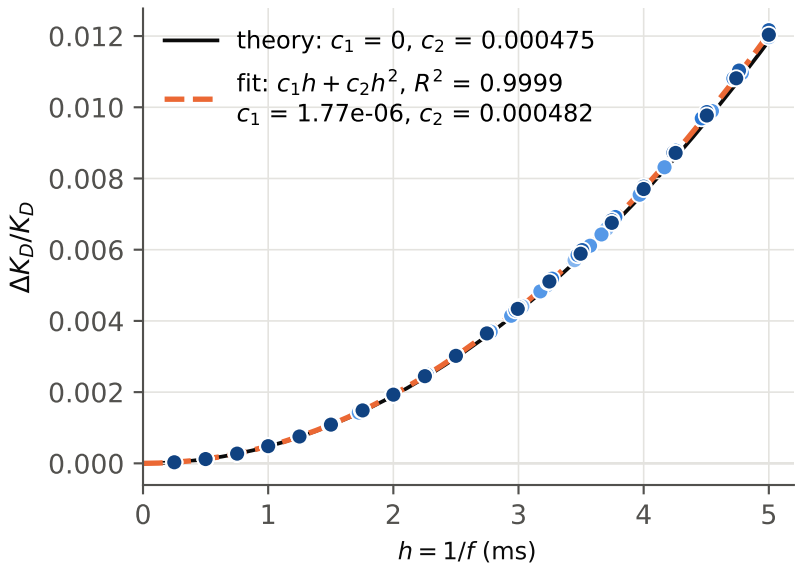
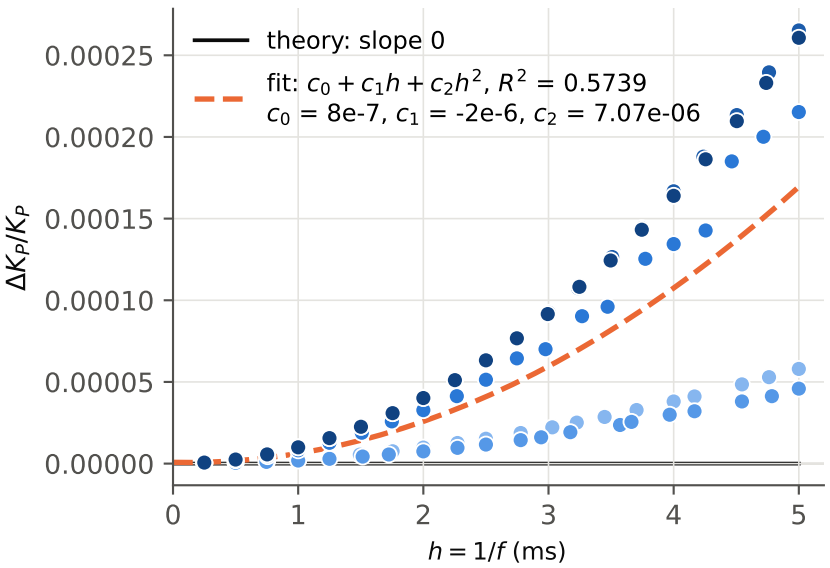
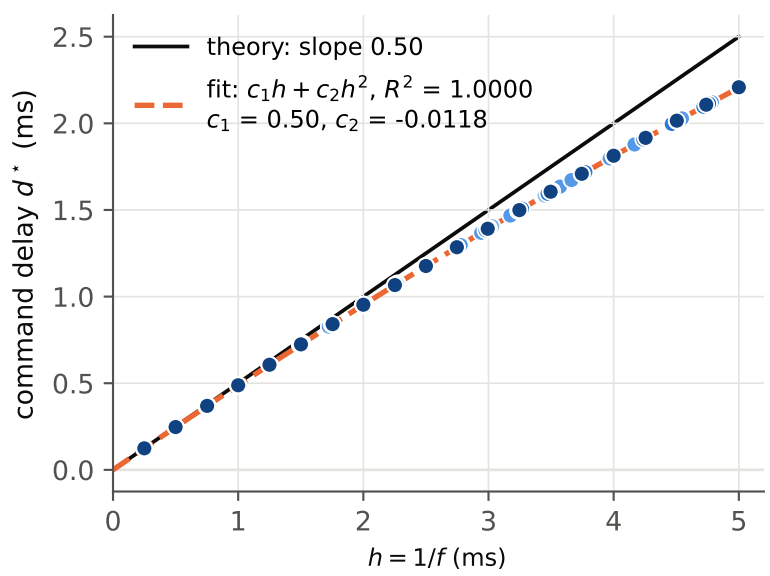
$f_{ctrl} \approx$ 10 Hz · $f_{ctrl} \approx$ 20 Hz · $f_{ctrl} \approx$ 50 Hz · $f_{ctrl} \approx$ 100 Hz · $f_{ctrl} \approx$ 200 Hz

A: Euler + explicit PD
fitted: J, K_P, K_D, d

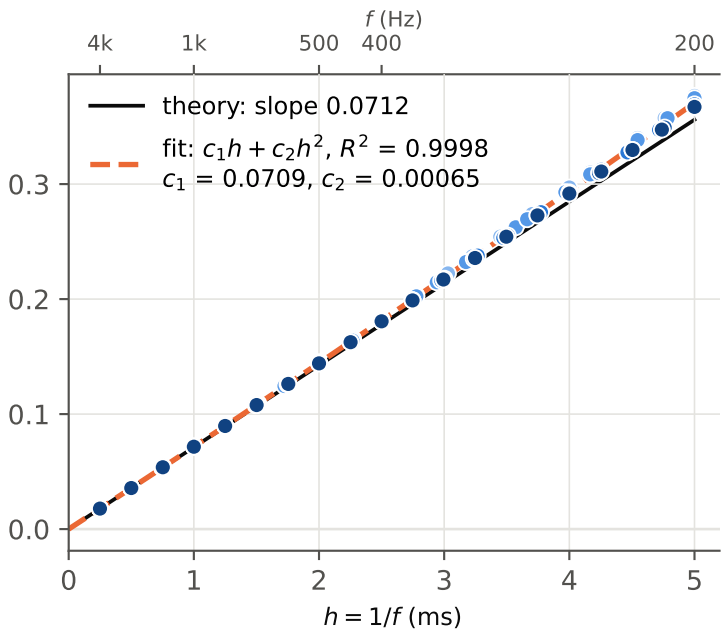


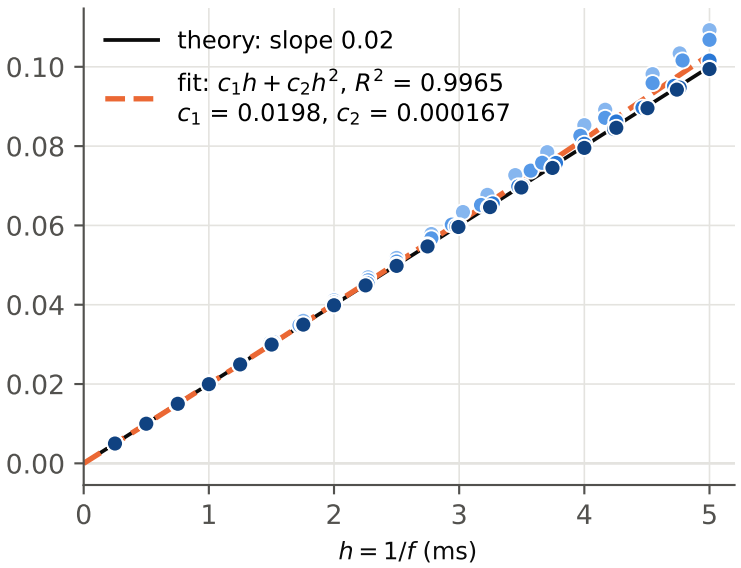


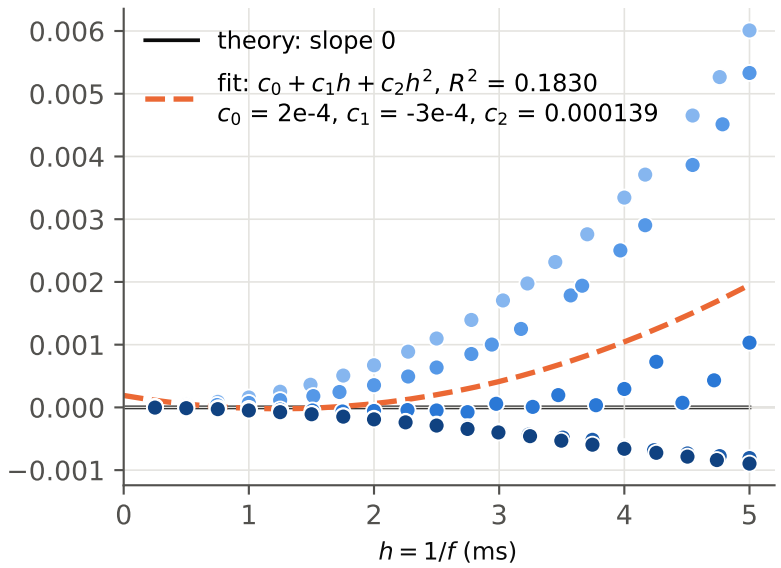


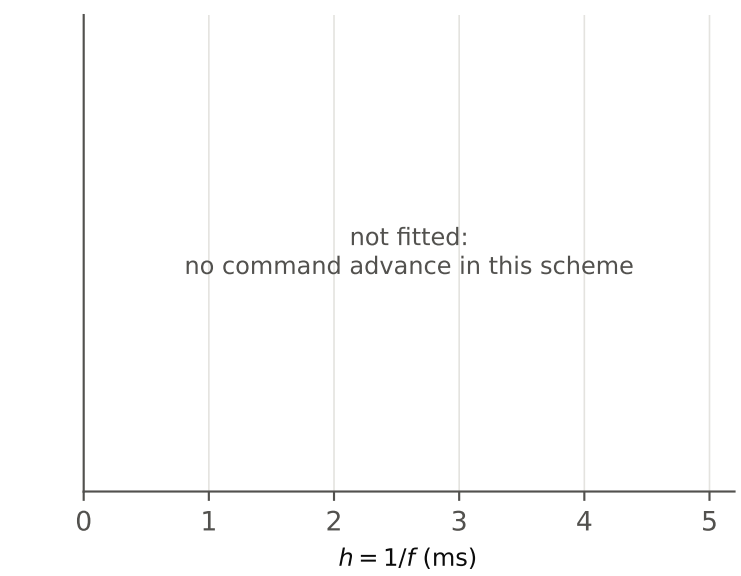


B: RK4 + explicit PD (torque held)
fitted: J, K_P, K_D

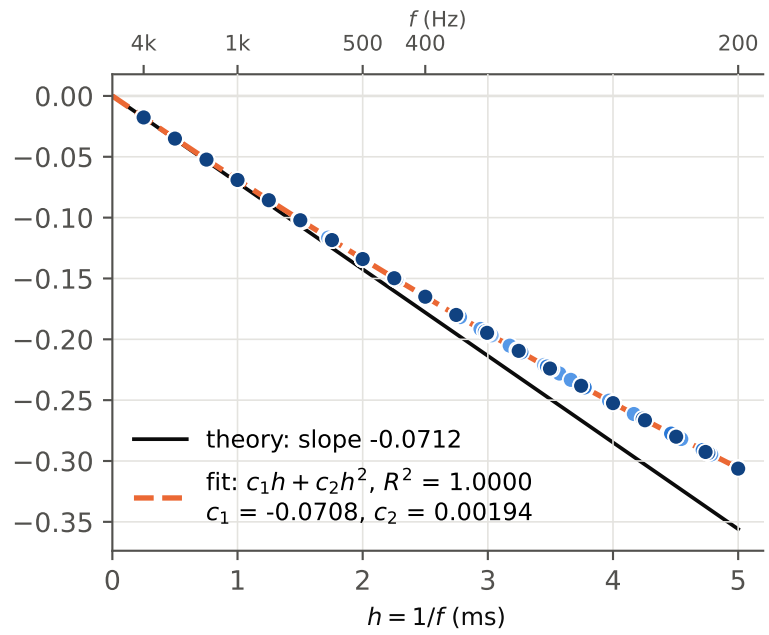


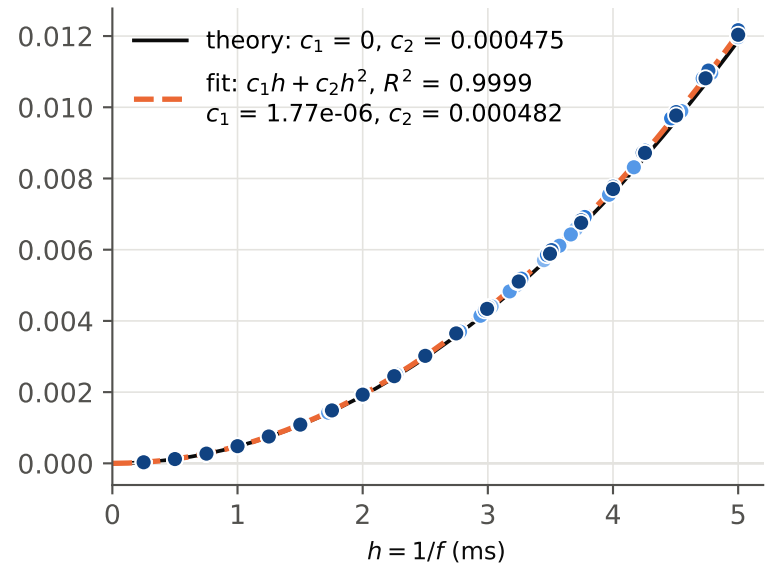


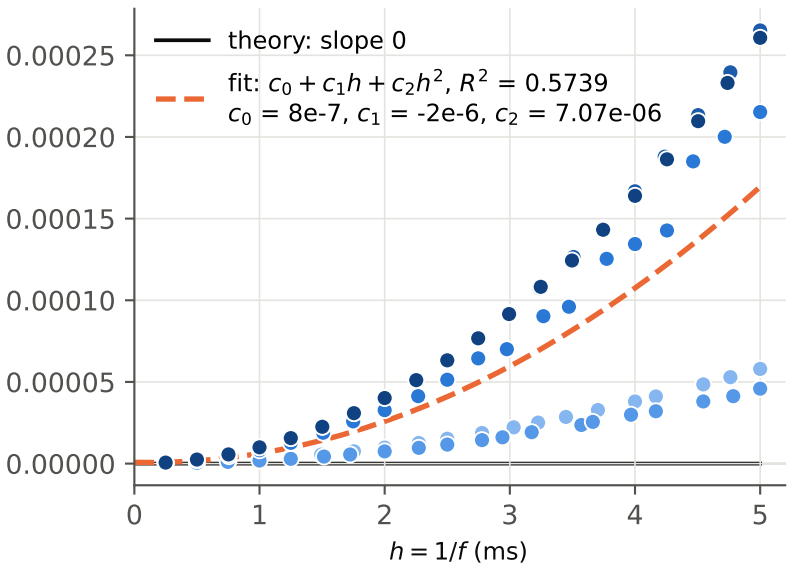


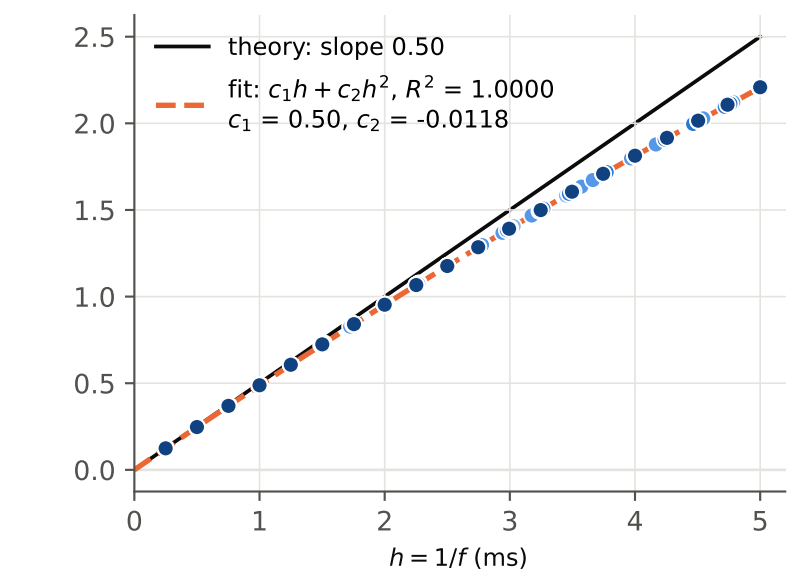


C: implicitfast + native PD
fitted: J, K_P, K_D, d









Target h = 0.25, 0.5, ..., 5 ms (f = 200-4000 Hz), realized with integer f_{ctrl} (Hz) and integer decimation, within 2.3% of the target.

y (solid), first order except the K_D row of A and C, which carries the derived h^2 counterterm $K_PK_D/12J$; fit (dashed), $c_1h + c_2h^2$ through the origin, so that c_1 is the measured first-order slope and c_2 absorbs the leading corre